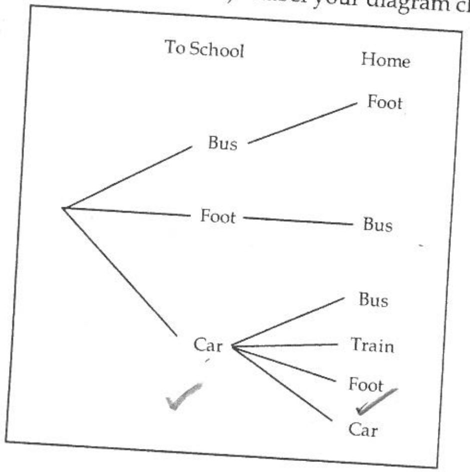
## Calculator Assumed

10. 10 marks:

Nicci can get to school either, by bus, on foot or by car. She can get home from school either, by bus, by train, on foot or by car. If she travels to school on foot, then she will take a bus home. If she takes a bus to school, then she will walk home.

(a) Draw a tree diagram to show the different ways Nicci can travel to school and back home (from school). Label your diagram clearly. ✓

Foot = Walk




outcomes
BF
FB
CB
CT
CF
CC ✓

(b) Assume that each combination of mode of travel to school and from school is equally likely.

(i) Find the probability that on a given day, she travels to school by car.

Probability = $\frac{4}{6}$ ✓✓

(ii) Find the probability that on a given day, she walks home from school.

Probability = $\frac{2}{6}$ ✓✓

(iii) Find the probability that on days when she gets to school by car, she takes a train home.

Probability = $\frac{1}{4}$ ✓✓





## Calculator Assumed

14. [12 marks:

4 for tree diagram.

Sharee's Café offers the following menu:

| | |
|---|---|
| Appetiser: | Spring-rolls, Chicken satay, Beef satay |
| Base: | Fried noodles, Fried rice, Steamed Rice |
| Main Dish: | Curried Fish, Curried Chicken, Sweet & Sour Chicken, Curried Lamb |
| Dessert : | Cendol, Ais-Kacang, Soya bean milk |

For a Set Meal, **each customer is allowed one choice from each of the four categories.**

(a) Explain clearly why there are 12 possible combinations of Base and Main Dish for a Set Meal.

There are 3 choices for the base.
There are 4 choices for the main dish.
Hence, together there are $3 \times 4 = 12$ combinations. ✓ ✓

(b) If Jon chose Chicken Satay, Fried Noodles and Curried Fish what is the probability that he also chose Ais-Kacang?

Prob. = $\frac{1}{3}$ ✓✓

(c) If Sharon chose Soya bean milk and Sweet & Sour Chicken, what is the probability that she chose Spring-rolls?

Prob. = $\frac{3}{9} = \frac{1}{3}$ ✓✓

(d) If in addition each of the three curried dishes came in mild, hot and super-hot, find the probability that Matt chose a hot curry dish given that he chose Beef satay with steamed rice and cendol.

Prob. = $\frac{3}{9} = \frac{1}{3}$ ✓✓